Power

File: power.kicad_sch

MCU

File: mcu.kicad_sch

Sensors

File: sensors.kicad_sch

Video

File: video.kicad_sch

Connectors

File: connectors.kicad_sch

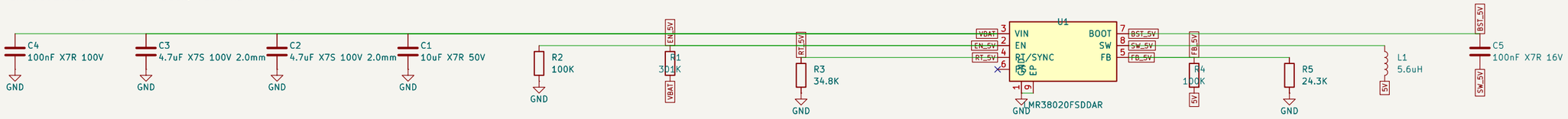
F405-FC

Sheet: /
File: F405FC.kicad_sch

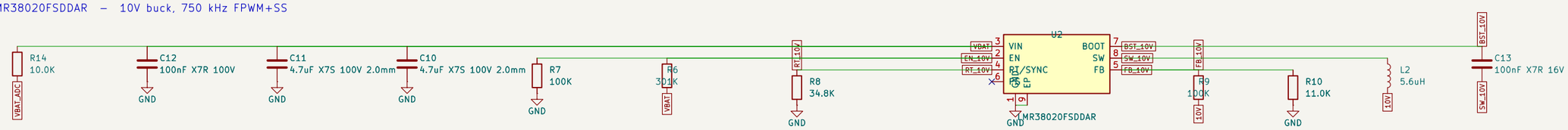
Title: F405-FC flight controller

Size: A4	Date:	**Rev: D**
KiCad E.D.A. 10.0.6		Id: 1/6

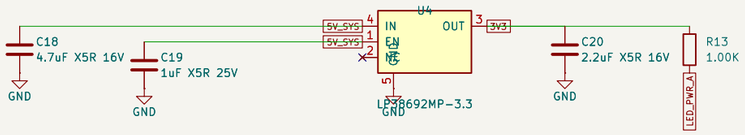
U1 LMR38020FSDDAR – 5V buck, 750 kHz FPWM+SS



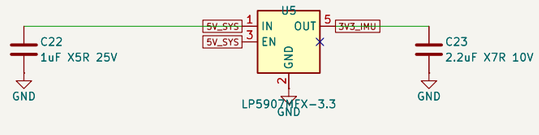
U2 LMR38020FSDDAR – 10V buck, 750 kHz FPWM+SS



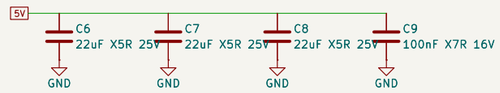
U4 LP38692MP-3.3 – 3.3 V 1 A LDO



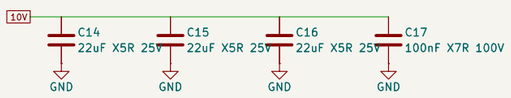
U5 LP5907MFX-3.3 – 3.3 V low-noise LDO for IMUs



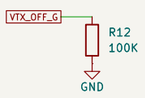
5V decoupling / bulk



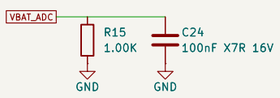
10V decoupling / bulk



VTX_OFF_G to GND

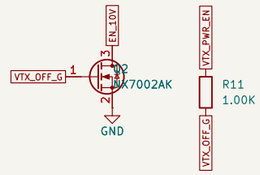


VBAT_ADC to GND











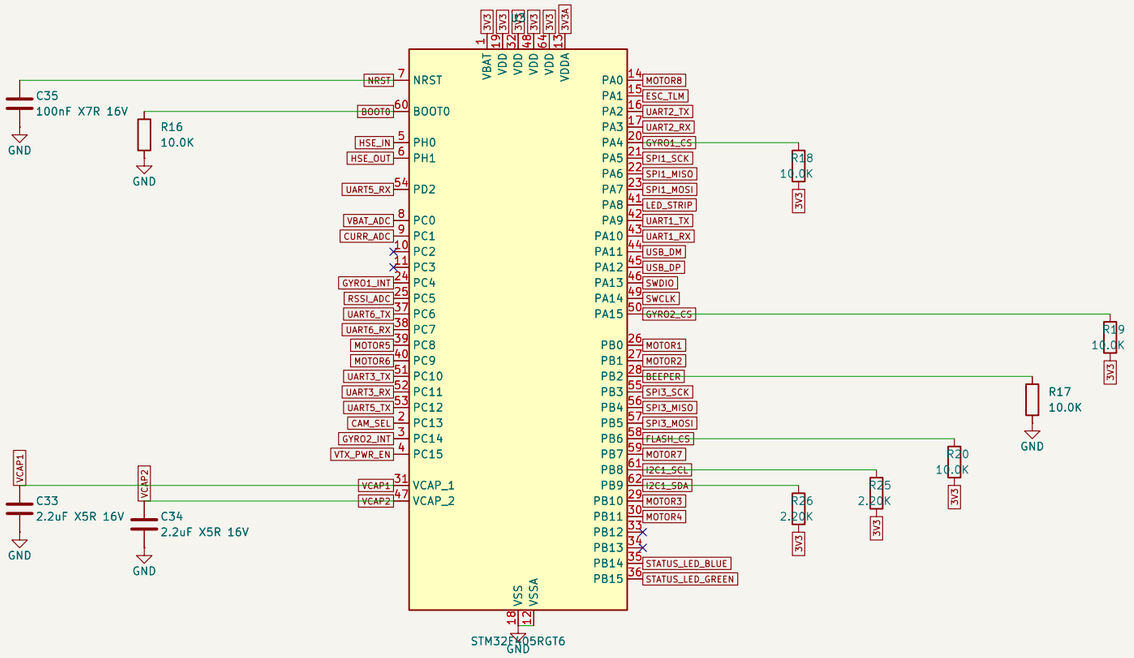




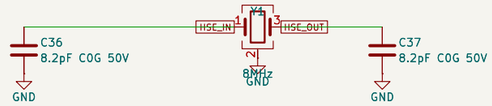
Power: see design/power.md rev B. U1/U2 LMR38020FSDDAR 750 kHz. U2 EN pulled low by Q2 when VTX_PWR_EN is high (VTX off).

F405-FC		
Sheet: /Power/		
File: power.kicad_sch		
Title: F405-FC flight controller – Power		
Size: A3	Date:	**Rev: D**
KiCad E.D.A. 10.0.6		Id: 2/6

U3 STM32F405RGT6 – MCU 168 MHz Cortex-M4F, LQFP-64

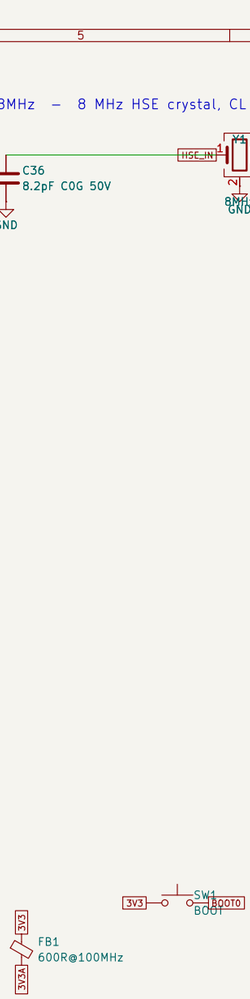


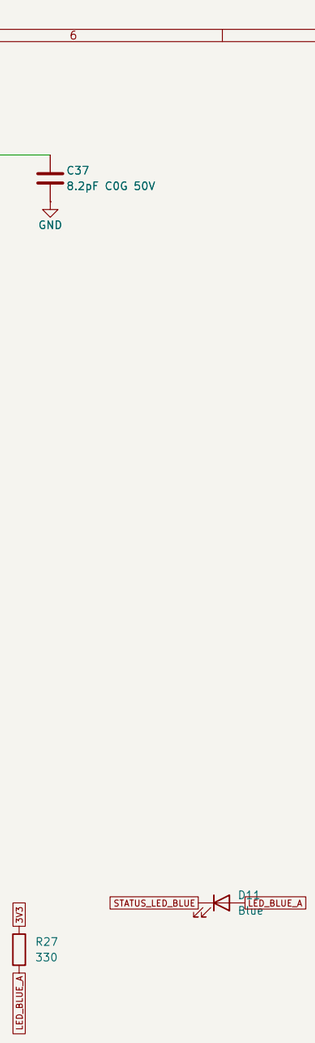
Y1 8MHz – 8 MHz HSE crystal, CL 8 pF



3V3 decoupling / bulk

3V3A decoupling / bulk

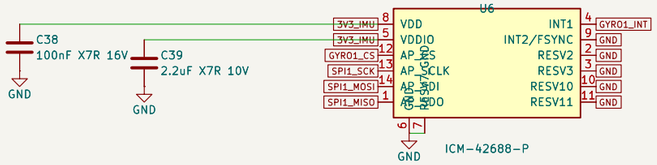




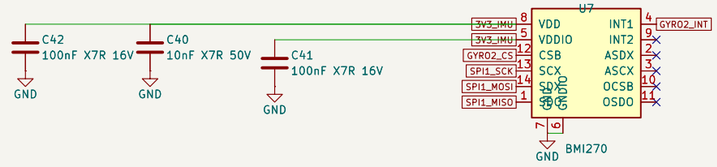
MCU: pin map design/pinmap.md rev C. PC2/PC3/PB12/PB13 unused (OSD removed, D-01).

F405-FC		
Sheet: /MCU/		
File: mcu.kicad_sch		
Title: F405-FC flight controller – MCU		
Size: A3	Date:	**Rev: D**
KiCad E.D.A. 10.0.6		Id: 3/6

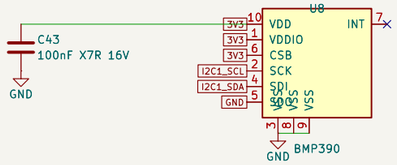
U6 ICM-42688-P – IMU 1 (gyro_1)



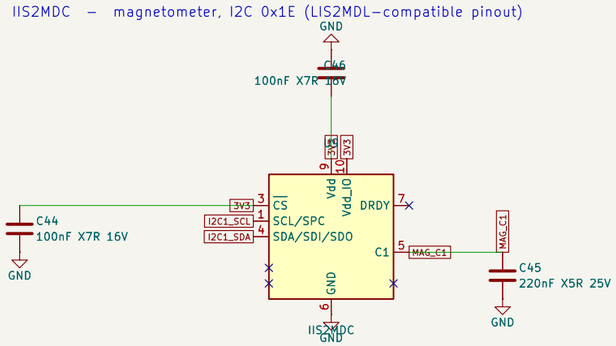
U7 BMI270 – IMU 2 (gyro_2)



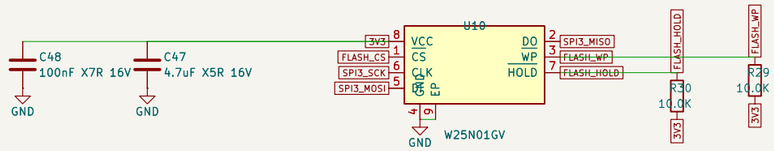
U8 BMP390 – baro, I2C 0x76



U9 IIS2MDC – magnetometer, I2C 0x1E (LIS2MDL-compatible pinout)



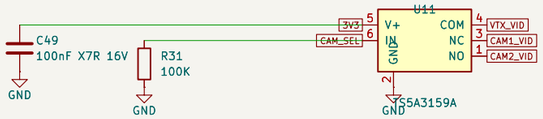
U10 W25N01GV – blackbox flash 1 Gbit



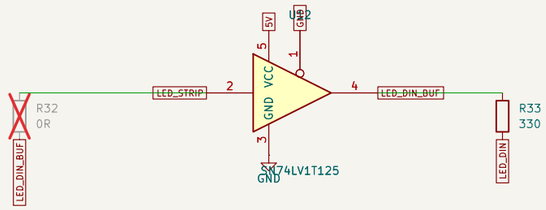
Sensors: IMUs on 3V3_IMU (LP5907), SPI1. Baro 0x76 + mag 0x1E on I2C1. Flash on SPI3.

F405-FC		
Sheet: /Sensors/ File: sensors.kicad_sch		
Title: F405-FC flight controller – Sensors		
Size: A3	Date:	**Rev: D**
KiCad E.D.A. 10.0.6		Id: 4/6

U11 TS5A3159A – camera select (low = CAM1)



U12 SN74LV1T125 – WS2812 3.3->5 V buffer



5V decoupling / bulk

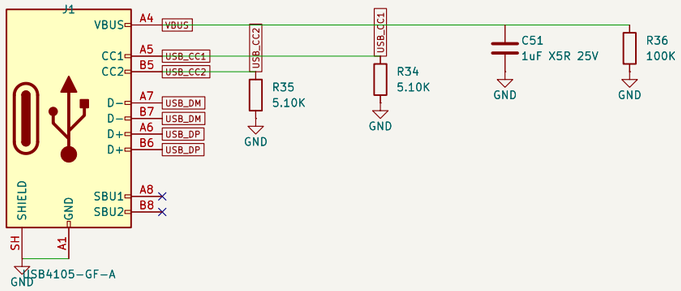




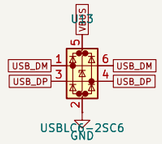
Video: analog camera switch only (no OSD chip, D–01); DC–coupled cameras only (signal must stay 0..3.3 V). LED buffer U12 from 5V; DNP bypass R32 only with U12 removed.

F405–FC		
Sheet: /Video/		
File: video.kicad_sch		
Title: F405–FC flight controller – Video		
Size: A3	Date:	**Rev: D**
KiCad E.D.A. 10.0.6		Id: 5/6

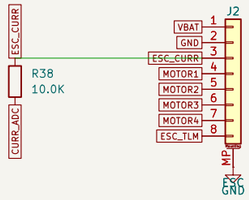
J1 USB4105-GF-A – USB-C (USB 2.0 FS)



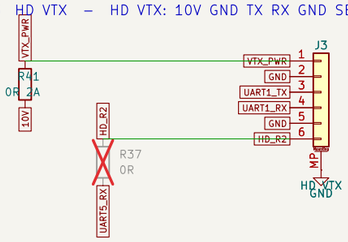
U13 USBLC6-2SC6 – USB ESD



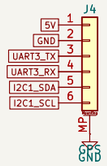
J2 ESC – ESC: VBAT GND CURR M1 M2 M3 M4 TLM



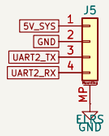
J3 HD VTX – HD VTX: 10V GND TX RX GND SBUS/R2



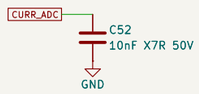
J4 GPS – GPS: 5V GND TX RX SDA SCL



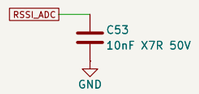
J5 ELRS – ELRS RX: 5V GND TX RX

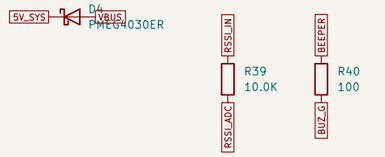


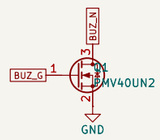
CURR_ADC decoupling / bulk

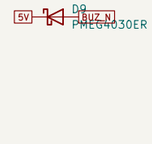


RSSI_ADC decoupling / bulk













Connectors: JST-SH. ESC pin order VBAT GND CURR M1 M2 M3 M4 TLM. HD VTX pin 6 (R2) NC unless R DNP fitted.

F405-FC		
Sheet: /Connectors/		
File: connectors.kicad_sch		
Title: F405-FC flight controller – Connectors		
Size: A3	Date:	**Rev: D**
KiCad E.D.A. 10.0.6		Id: 6/6